

# Isaac Mubiru

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Kampala, Uganda (UTC+3) — Remote worldwide

## PROFESSIONAL SUMMARY

Agentic engineer on a real backend foundation: 6 years of Node.js/TypeScript production systems — databases, integrations, APIs — now applied to orchestrating AI agents critically, not prompt-and-hope. Built and open-sourced a multi-step agentic pipeline on Claude Code ([github.com/zac-09](https://github.com/zac-09)): Playwright ground-truth verification, batch workers, integrity checks, documented failure handling; 100+ runs. Shipped conversational AI in production (WhatsApp Business API + OpenAI + Google Vision OCR). Foundation work: led a zero-downtime migration of 20+ apps to Firebase/GCP with a real-time two-way Pub/Sub sync; architected Kafka/Kubernetes microservices. Remote from Kampala (UTC+3), contract-experienced.

## CORE COMPETENCIES

Agentic Workflow Orchestration

Claude Code · Cursor · TypeScript

Node.js Backend Foundation

LLM Production Systems

API & Data Integrations

System Architecture

## WORK EXPERIENCE

**MTailor** (US Remote)

Sep 2022 – Present

Software Engineer — reporting directly to the CTO

- Shipped **Express Shipping end-to-end** (concept → pricing → backend → ops handoff), generating an additional **\$40 revenue per order**
- Built a **3D visualisation feature** for customers using video overlay and ffmpeg, **increasing buyer conversion**
- Led the zero-downtime migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP
- Implemented real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub
- Migrated file storage from Amazon S3 to GCS via a Python script; moved all services off AWS, saving **\$5,000/month**
- Migrated a Webflow website from a Parse backend to Firebase

**MTailor** (US Remote)

Mar 2022 – Jul 2022

Contractor (Backend Migration)

- Hired via Upwork onto an unfamiliar stack; built MongoDB→Firestore migration scripts with complex processing logic
- Wrote Firebase SDK documentation (Node.js and Python) other engineers onboarded from; trained the Ops team — converted to full-time and given the company-wide migration lead within months

**CodeBits** (Uganda)

Jan 2020 – Jul 2021

Led a team of 4 developers building legal tech for NGOs end-to-end

- Owned client relationships and delivery for FIDA Uganda and LASPNET — requirements, architecture, build, ship
- Architected an event-driven microservices backend using **Apache Kafka, Docker, and Kubernetes**
- Built a **USSD service** for legal aid over feature phones, communicating with a Node.js backend via gRPC — meeting rural users on the infrastructure they actually have
- Developed a cross-platform mobile app using **React Native and Expo** (iOS & Android) with Firebase push notifications

**Dr Wealth** (Singapore Remote)

Aug 2021 – Jan 2022

Software Engineer

- Led optimisation of a PWA delivering real-time stock prices; built jobs querying 2M+ Firestore records from Morningstar APIs
- Extended backend APIs with Firebase Cloud Functions and Express.js; built responsive React/Tailwind UI pages

**Mind2matter** (US Remote)

Jan 2022 – Jul 2022

Software Engineer (Full-Stack)

- Built backends for DeFi applications using Web3 and Node.js; delivered pixel-accurate React UIs from Figma under tight timelines

## PROJECTS

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### career-ops — Multi-Step Agentic Pipeline (open source)

Agentic system on Claude Code: Playwright ground-truth verification (snapshots over cached search), batch evaluation workers, dedup/merge integrity scripts, automated document generation. 100+ production runs with documented failure autopsies.

Claude Code · Playwright · Node.js · YAML

### Healthcare WhatsApp Chatbot (LLM in production)

Conversational AI over the WhatsApp Business API: OpenAI dialogue + Google Vision prescription OCR under third-party platform constraints.

Node.js · OpenAI · Google Vision · WhatsApp Business API

### Marketplace Recommendation Engine

Collaborative + content-based filtering engine in MongoDB for service-provider matching in a live marketplace.

Node.js · MongoDB

## EDUCATION

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BSc Computer Engineering — [Makerere University](#), Kampala, Uganda

2018 – 2022

## SKILLS

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**Proficient:** Node.js, TypeScript/JavaScript, React, Firebase, GCP, Python, MongoDB, SQL, Git, Linux    **Intermediate:** Go, AWS, Rust, React Native

**Infrastructure:** Kubernetes, Docker, Kafka, gRPC, Pub/Sub    **Languages:** English, Kiswahili